

House Price Prediction

A supervised regression pipeline covering EDA, preprocessing, feature engineering, and model interpretation.

- **PRESENTED BY**
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- **TECH STACK**
Python, XGBoost, SHAP, Streamlit



Project Objectives

THE GOAL: PREDICT RESIDENTIAL PROPERTY SALE PRICES ACCURATELY USING STRUCTURAL AND QUALITATIVE HOUSING ATTRIBUTES.



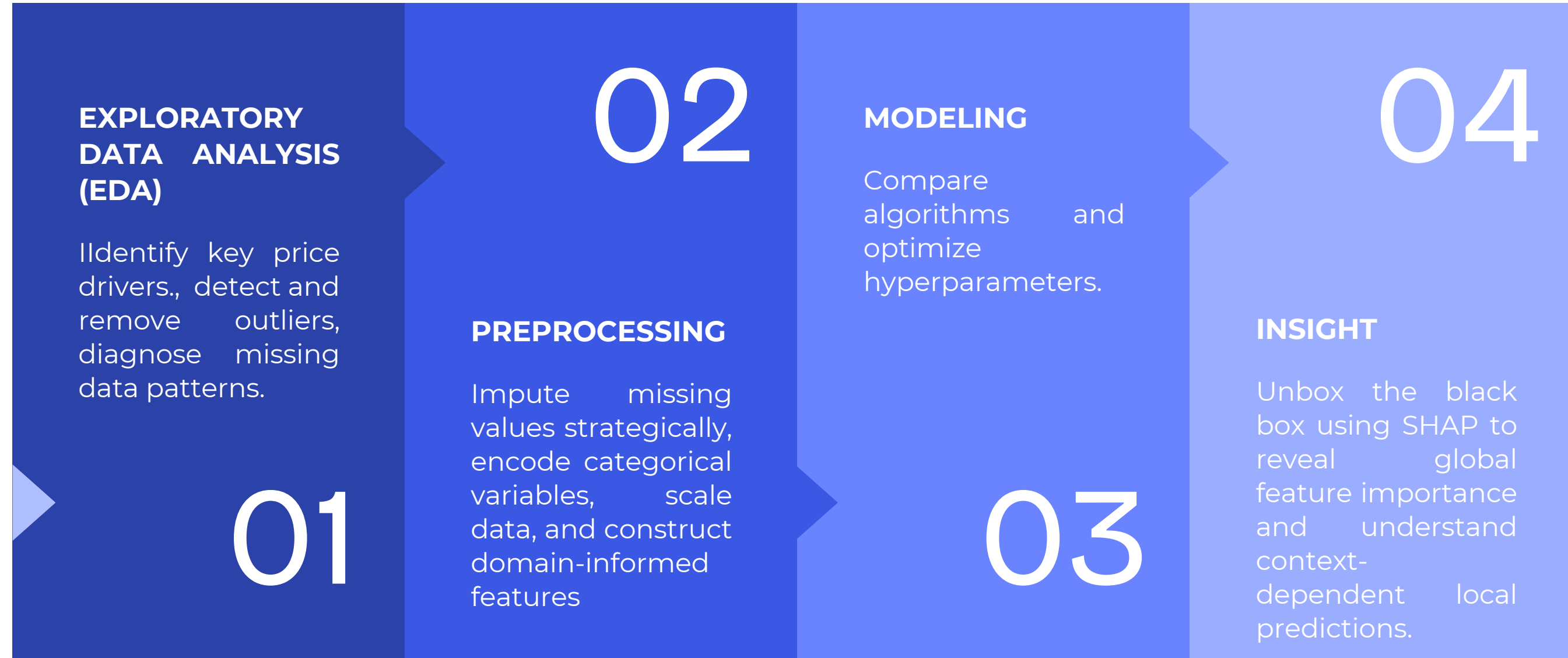
Why It Matters

- Supports data-driven house valuation.
- Identifies key price-driving factors.
- Provides a benchmark for regression models.
- Enhances model interpretability with SHAP.



PINELINE

**THE GOAL: PREDICT RESIDENTIAL PROPERTY SALE PRICES
ACCURATELY USING STRUCTURAL AND QUALITATIVE HOUSING
ATTRIBUTES.**



01



LINER REGRESSION

Finds the best-fitting straight line through data points to minimize prediction errors, assuming a direct linear relationship between features and house prices.

02



RANDOM FOREST

An ensemble method that builds multiple independent decision trees from random data subsets and averages their predictions to ensure stability and prevent overfitting.

03



XGBOOST

Builds decision trees sequentially, where each new tree focuses on correcting the errors of the previous ones, delivering exceptionally high accuracy for tabular data.

04



MODEL COMPARISON

Evaluates algorithms using RMSE, MAE, and R^2 metrics.

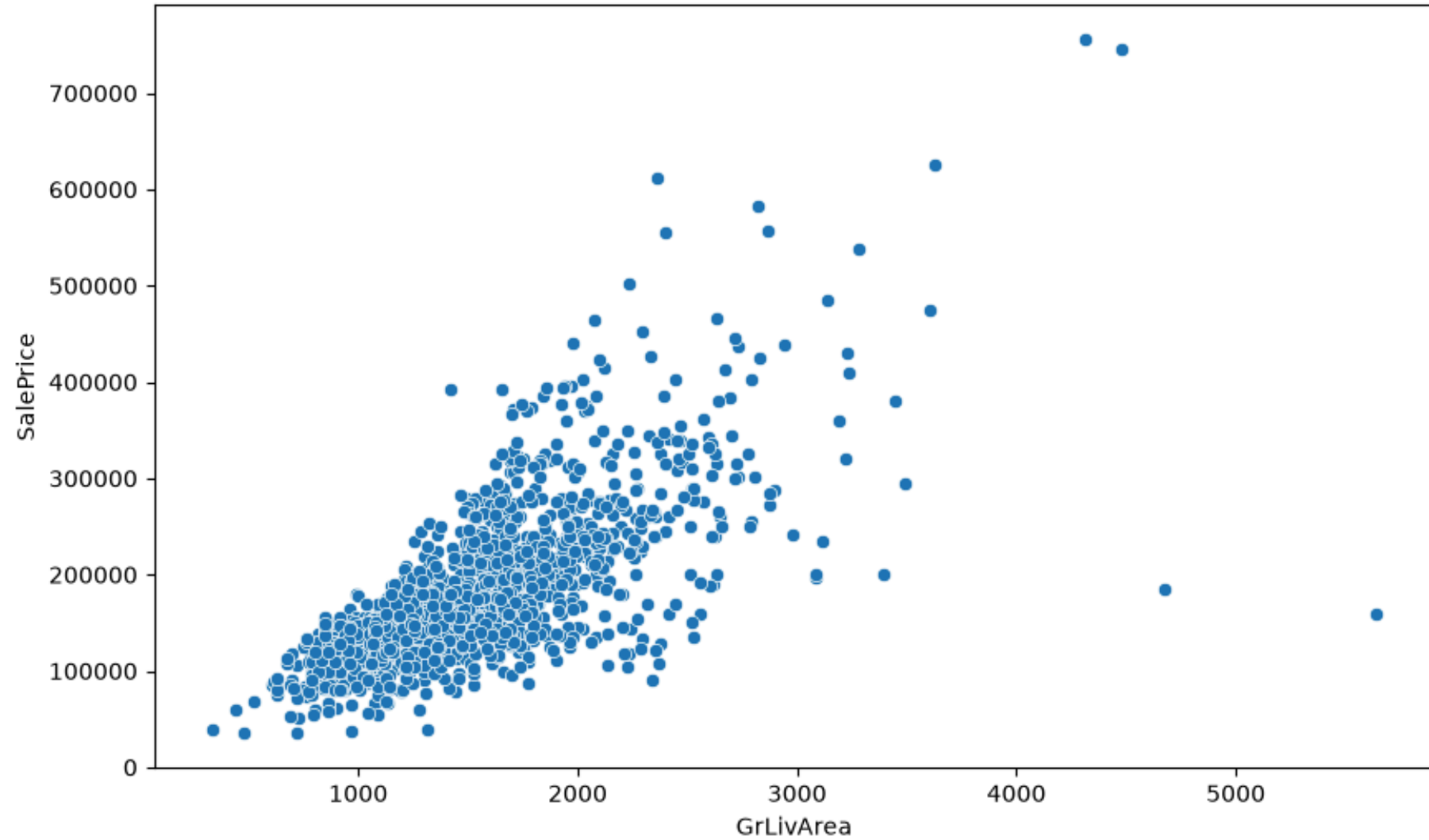
03

Research Methodology

Utilizing Python, systematic data preprocessing, regression modeling, and SHAP analysis for robust evaluation.

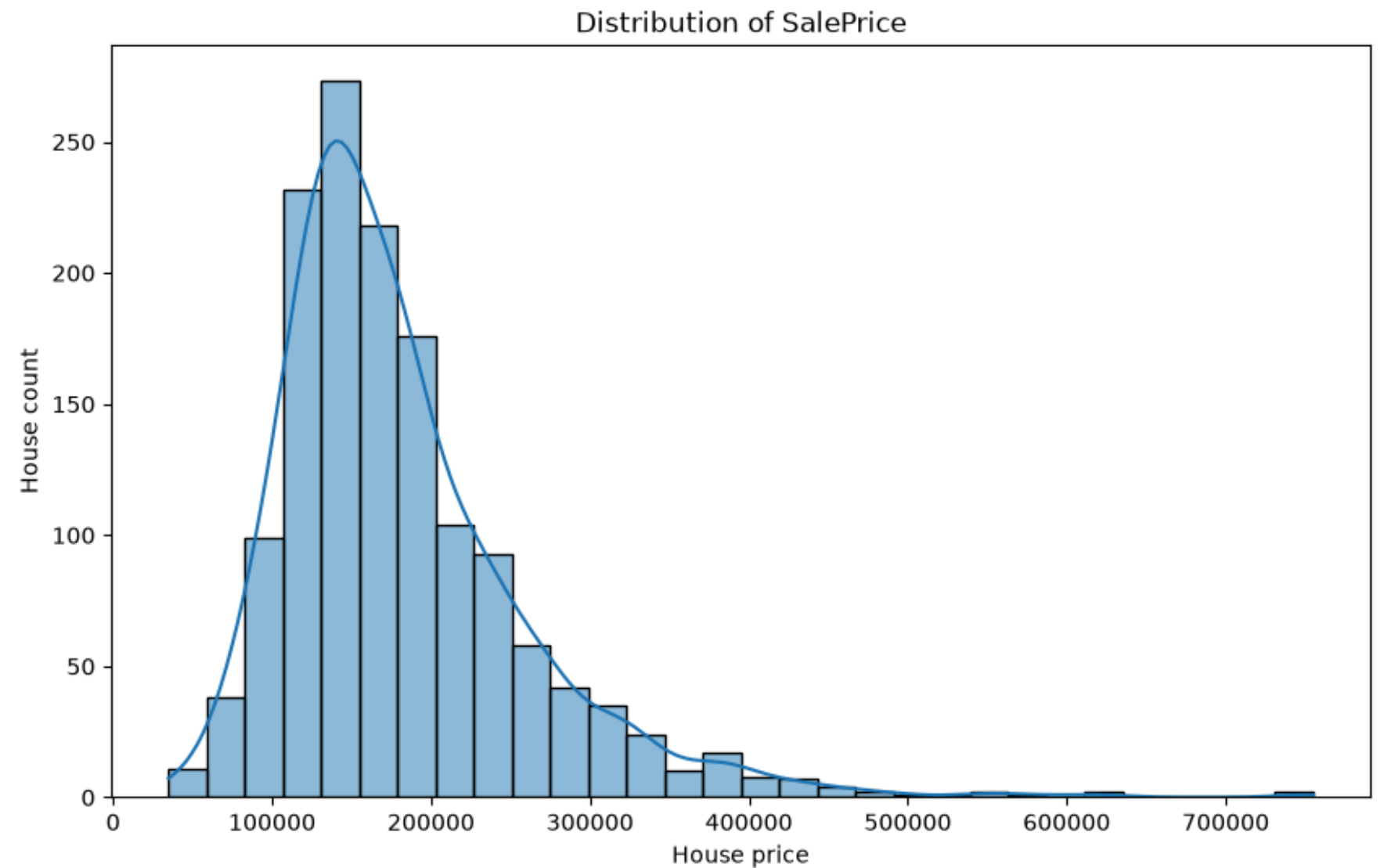
Exploratory Data Analysis (EDA)

04



The trend shows that as OverallQual increases, SalePrice also increases.

There are two clear outliers, with areas of approximately 4,700 and 5,700 square feet.



Min: 34900
Max: 755000
Mean: 180921.19
Median: 163000.0
Standard Deviation: 79442.5

It is noticeable that the histogram is right-skewed (mean > median).

PREPROCESSING

05

1. MISSING VALUES

“None” for structurally-absent categorical features; 0 for absent numeric features; median / mode for genuine missingness.

2. CROSS-REFERENCE IMPUTATION

GarageYrBlt filled from YearBuilt when no garage records exist.

3. ORDINAL ENCODING

2. Cross-reference imputation

GarageYrBlt filled from YearBuilt when no garage records exist.

4. ONE-HOT ENCODING

Nominal categories expanded feature count from 81 to 235 columns.

5. STANDARD SCALER

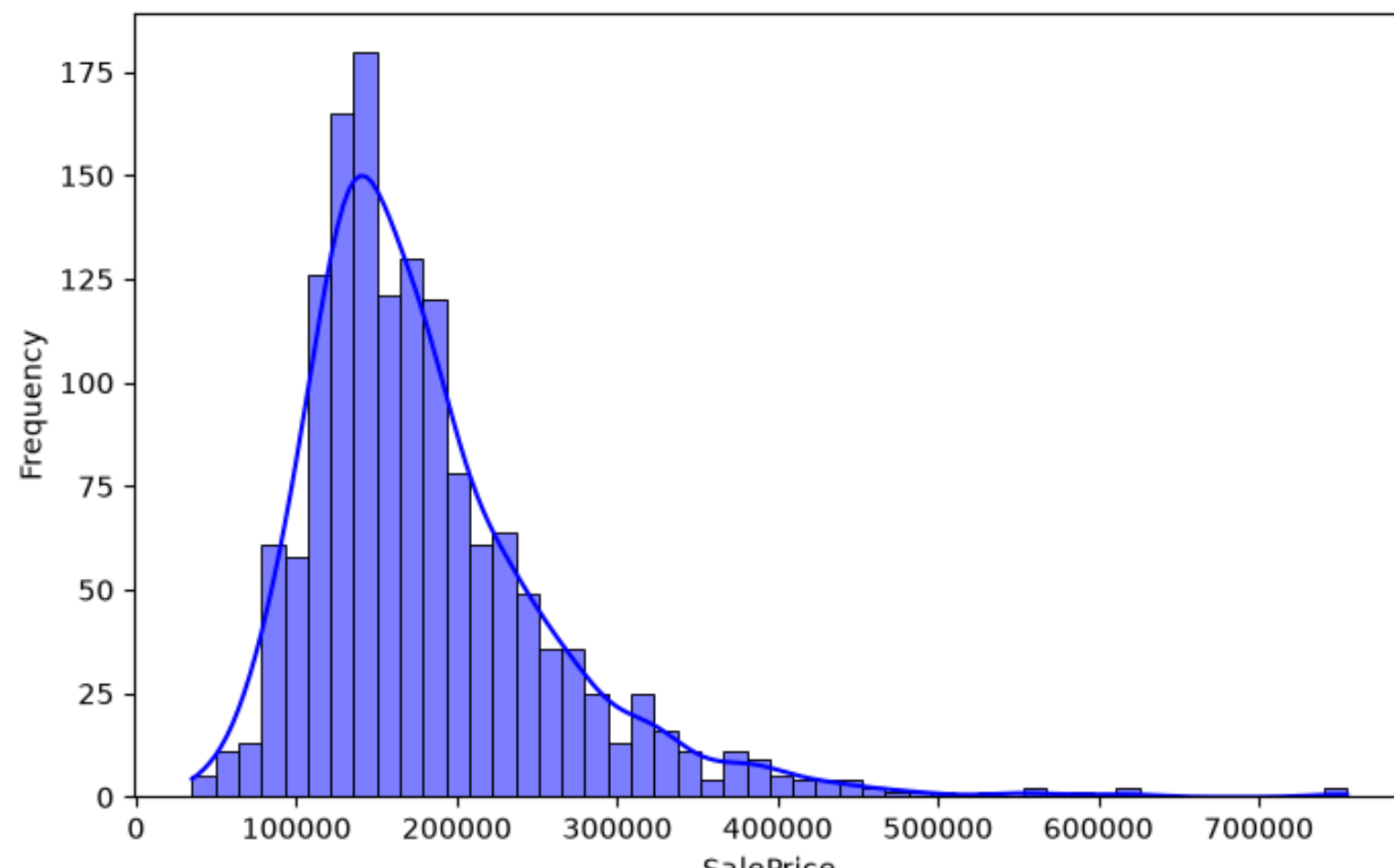
Fit on train only, transform on test to prevent data leakage.

6. TARGET LOG-TRANSFORM

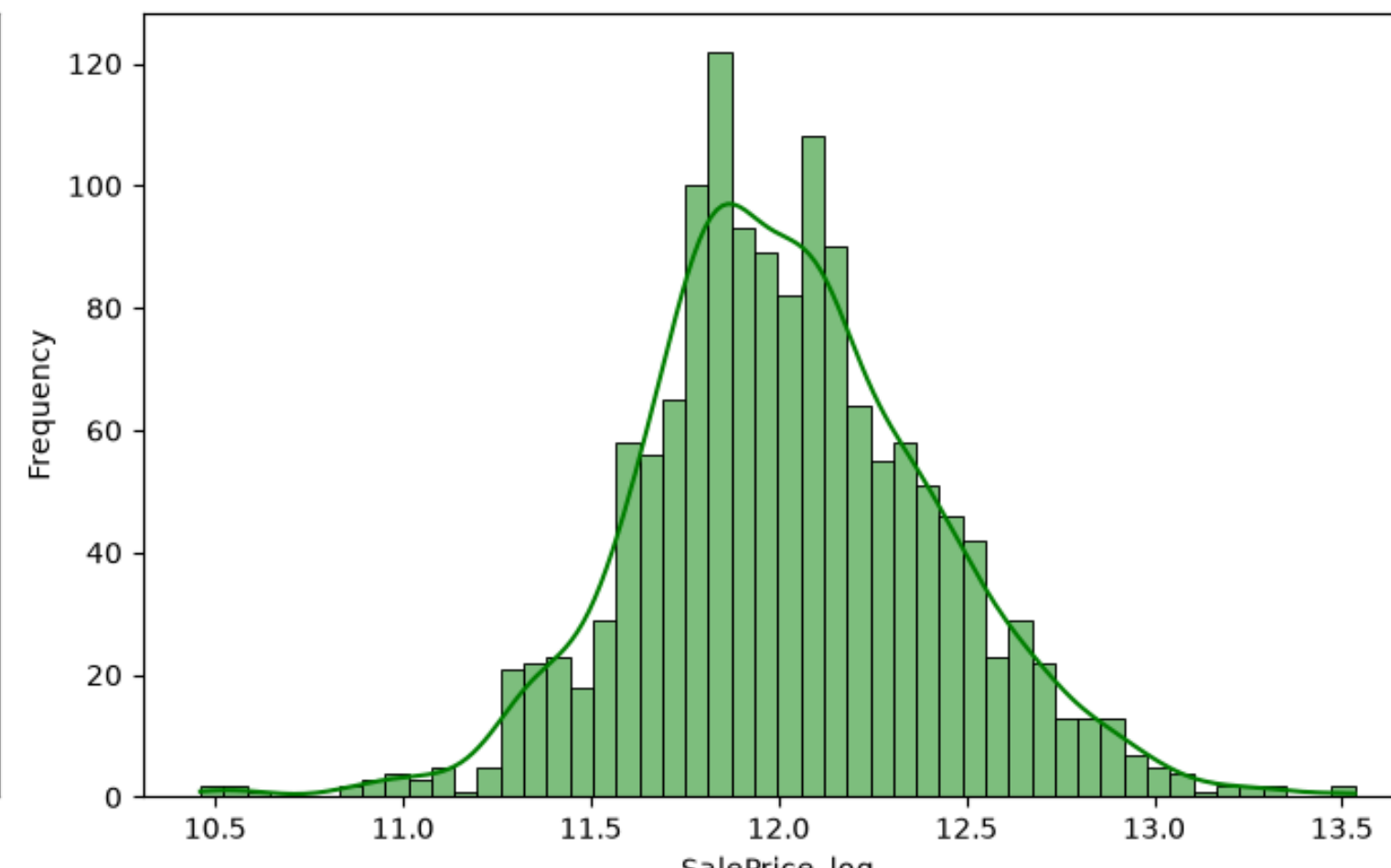
$\text{SalePrice_log} = \log_{1p}(\text{SalePrice})$ reduces skew and stabilizes linear baseline

$\text{np.log1p}(x)$ helps reduce the skewness of the data, and the skewness of SalePrice after the log transformation ($\text{SalePrice_skew_log}$) is much closer to 0 than the original SalePrice_skew

Original SalePrice Distribution
Skewness: 1.88



SalePrice_log Distribution (log1p)
Skewness: 0.12



MODELING

BEFORE LOG-TRANSFORM & FEATURE ENGINEERINGESS PERFORMANCE

01

BASELINE COMPARISON

Linear Reg (R^2 0.859)
< Random Forest (R^2 0.891) < XGBoost
default (R^2 0.901).

02

INITIAL TUNING

Optimized XGBoost via
GridSearchCV. Best baseline
params: learning_rate=0.05,
max_depth=3,
n_estimators=300.

03

INTERMEDIATE METRICS

The tuned baseline
model achieved:
RMSE 20,232.65, MAE
14,772.80, and an R^2 of
0.9259

04

RAW FEATURE INSIGHTS

OverallQual strongly
correlates. GrLivArea's
importance dropped due to
multicollinearity.

MODELING

AFTER LOG-TRANSFORM & FEATURE ENGINEERING

01

BASELINE COMPARISON

Random Forest (R^2 0.895) < XGBoost default (R^2 0.899) < Linear Reg (R^2 0.912).

02

FINAL TUNING

GridSearchCV optimized XGBoost with the new best params: `learning_rate=0.1`, `max_depth=3`, `n_estimators=300`.

03

FINAL METRICS

The final tuned best_XGBoost model achieved the highest scores: RMSE 19,650.11, MAE 13,787.39, and an R^2 of 0.9301.

04

ENGINEERING IMPACT

Applying log-transformation and domain-knowledge features successfully boosted the R^2 from 0.9259 up to 0.9301.

UNBOXING THE BLACK BOX WITH SHAP ANALYSIS

- 1. TotalSF (engineered feature) ranks #1, even ahead of OverallQual — high values (red) push predictions up, low values (blue) push down. Confirms feature engineering added real predictive value, not just theoretical.
- 2. OverallQual remains a dominant driver (#2) — clean separation between high quality (red, pushes price up) and low quality (blue, pushes price down), little overlap in the middle.
- 3. TotalBath (engineered feature) also lands in top 3, reinforcing that combining raw bath-related columns into one feature helped the model.
- 4. HouseAge behaves as expected: newer houses (blue = low age) push price up, older houses (red = high age) push price down

